Pentagons for the Kites

How one idea for the DICTO Ogdoecontahedron grew: what it settled, what it did not, what it opened, and the shapes found and re-found on the way



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5 November 2026 · Licence CC BY-SA 4.0 · Data, scripts and checks: https://github.com/DICTOR-Master/kaleidohedra (branch research, studies/2026-11-05-og-pentagons).

DRAFT for DICTO's reading. Not published. Delete this box at publication. Working title (DICTO to choose title and number). Names for the new pieces in §5 are DICTO's to give. Before release: exact (not float) Dehn check; closed form of the equal-edge solid; a check of the near-miss catalogues; the gyrated-octet twin-layer pieces (being identified); prior-art search on the cut-OG kits. OG #31 Version 5 will then carry one paragraph pointing here; Bull-horns, Golden Bodhi Pyrochlore and the gyrated-octet paper a one-line pointer each, batched.

The DICTO Ogdoecontahedron is a ball of 20 hexagons and 60 kites, the kites gathered in twelve five-pointed stars. DICTO proposed putting a pentagon in the middle of each star. One flat cut at each of the twelve star points does it, and what is left turns out to be hiding a famous shape: its corners sit exactly at the midpoints of the edges of a football, the shape of the carbon molecule C_{60}. From there the idea kept growing.

Abstract. DICTO proposed replacing the five-kite stars of the DICTO Ogdoecontahedron (OG) with pentagons. One plane cut at each of the twelve five-fold corners, through the side corners of its five kites, removes exactly the twelve corners that lie off the OG's sphere and nothing else. **Theorem.** The convex hull of the OG's 90 sphere corners is the rectified truncated icosahedron: the 90 corners are the edge midpoints of a regular truncated icosahedron. Following DICTO's further proposals we find: an all-equal-edge version (equilateral triangles, regular pentagons, hexagons with alternating angles); hexagons as truncated triangles, possible only in a chiral, left and right family; a nonzero Dehn invariant (float screen) and a best lattice packing of 0.776244; a kit of four shapes and a left/right pair for the octet truss, built on a pentagonal connector across every octahedron edge; and DICTO's Bull-horns OG adaptor adapted to the cut solid, so that two shapes fill space. The gyrated octet is open. The solid itself is known; the relation to the OG, the chiral family and the kits were not found in the sources checked.

1. How the idea evolved

The OG's twenty regular hexagons surround twelve stars of five kites. DICTO proposed setting a pentagon in the middle of each star (§2). The cut that does it revealed the football (§2). DICTO then proposed making the triangles more regular; with all edges equal the triangles become equilateral and the hexagons give way (§2). He proposed instead making the hexagons truncated triangles; that works only when the solid gives up its mirrors and twists, left or right (§3). He proposed trying the cut solid where the OG already lives: the octet truss and its gyrated form, and the body-centred cubic double network with his Bull-horns adaptor (§4).

2. What it settled

Theorem 1. Cut each of the OG's twelve five-fold corners by the plane through the side corners of its five kites. Nothing else is cut, each star becomes a regular pentagon, each kite an isosceles triangle, the hexagons are unchanged, and the solid that remains is the convex hull of the OG's 90 sphere corners (radius 2/3). These 90 points are the edge midpoints of a regular truncated icosahedron, so the solid is the rectified truncated icosahedron.

Check. The twelve removed corners are exactly the OG's twelve corners off the sphere; 98.35% of the volume remains. The triangles have base 2/9 (on a pentagon) and sides 0.2379 (on hexagons), with angles 55.7°, 62.2°, 62.2°. The ratio of the two edge lengths is 1.07047 = sin 60° / sin 54°, the ratio for the edge midpoints of a regular truncated icosahedron, and measured directly every corner lies on such a midpoint to within 10^{-10} (Figures 1 and 2). The truncated icosahedron is the shape of the C_{60} molecule (Kroto et al. 1985), so the 90 points sit at its bond centres.

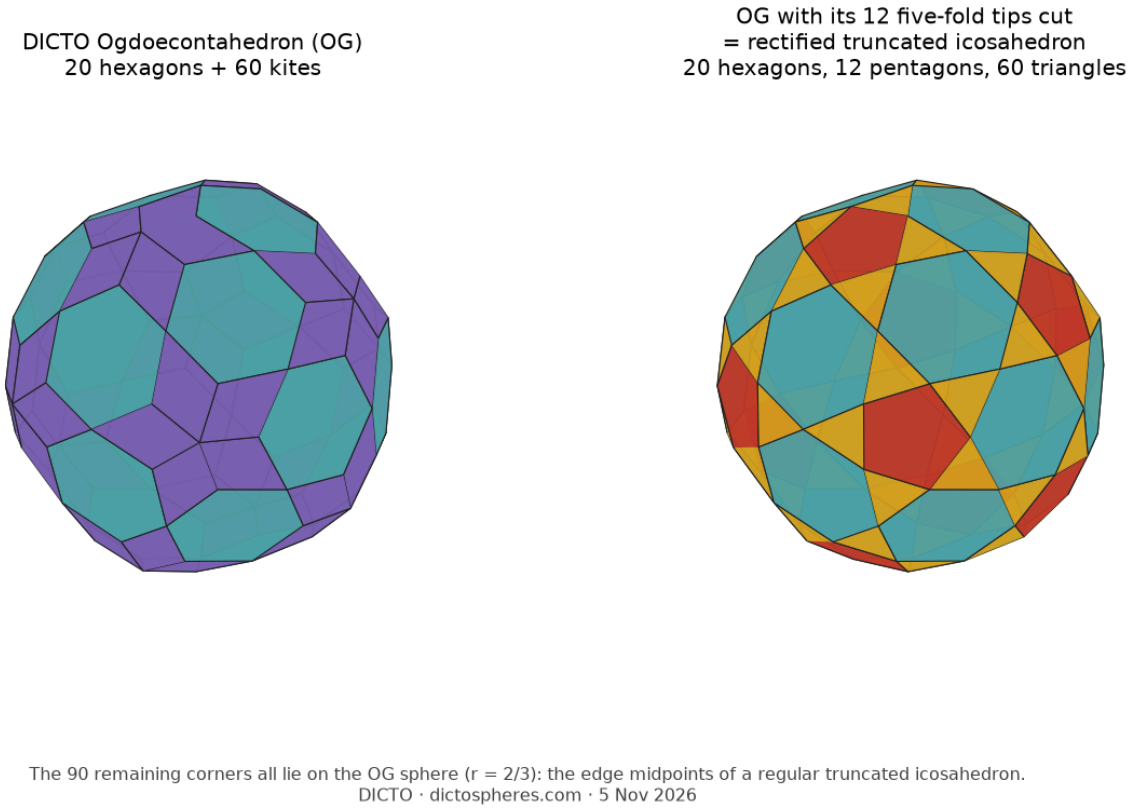


Figure 1. Left: the Ogdoecontahedron (20 hexagons, 60 kites). Right: the twelve star points cut off; 20 hexagons, 12 pentagons, 60 triangles, all 90 corners on one sphere.

All edges equal. The same arrangement can be built with all 180 edges equal and all faces flat, keeping full icosahedral symmetry: equilateral triangles, regular pentagons, and hexagons with all sides equal whose angles alternate 124.72° and 115.28° (numerical, residual 10^{-6}). Its corners then lie on two spheres (radius ratio 1.0058). Regular hexagons and equilateral triangles never occur together here, which is why this arrangement is counted among the near-misses (Figure 2, right).

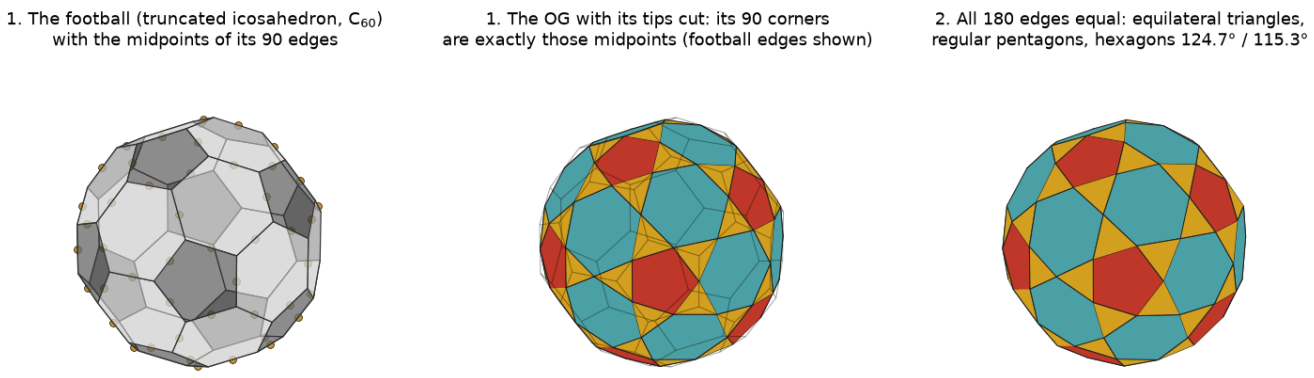


Figure 2. Left: the truncated icosahedron (the C_{60} football) with the midpoints of its 90 edges. Middle: the cut OG drawn with the football's edges; its corners are those midpoints. Right: the all-equal-edge version.

Space filling and packing. The Dehn invariant of the cut solid is nonzero (float screen, calibrated on the cube and rhombic dodecahedron), so it does not fill space alone. A lattice-packing search (299 local searches; verified over lattice coefficients up to 4) gives a best density of 0.776244, against 0.789260 for the OG; each copy touches 12 neighbours, 10 on hexagons and 2 on triangles.

3. Hexagons as truncated triangles: a left and a right

DICTO proposed making the hexagons truncated triangles, every angle 120° and the sides alternating long and short. With the mirrors kept this is impossible, since a mirror through each hexagon forces its six sides equal. Keeping only the 60 rotations, the solid on one sphere twists through a one-parameter family in which all twenty hexagons are flat truncated triangles, the pentagons stay regular, and the solid stays convex at least until the sides reach 0.114 and 0.343 (numerical, faces flat to 10^{-9}). Each member comes as a left and a right (Figure 3). The triangles become scalene, so truncated-triangle hexagons and equilateral triangles never occur together.

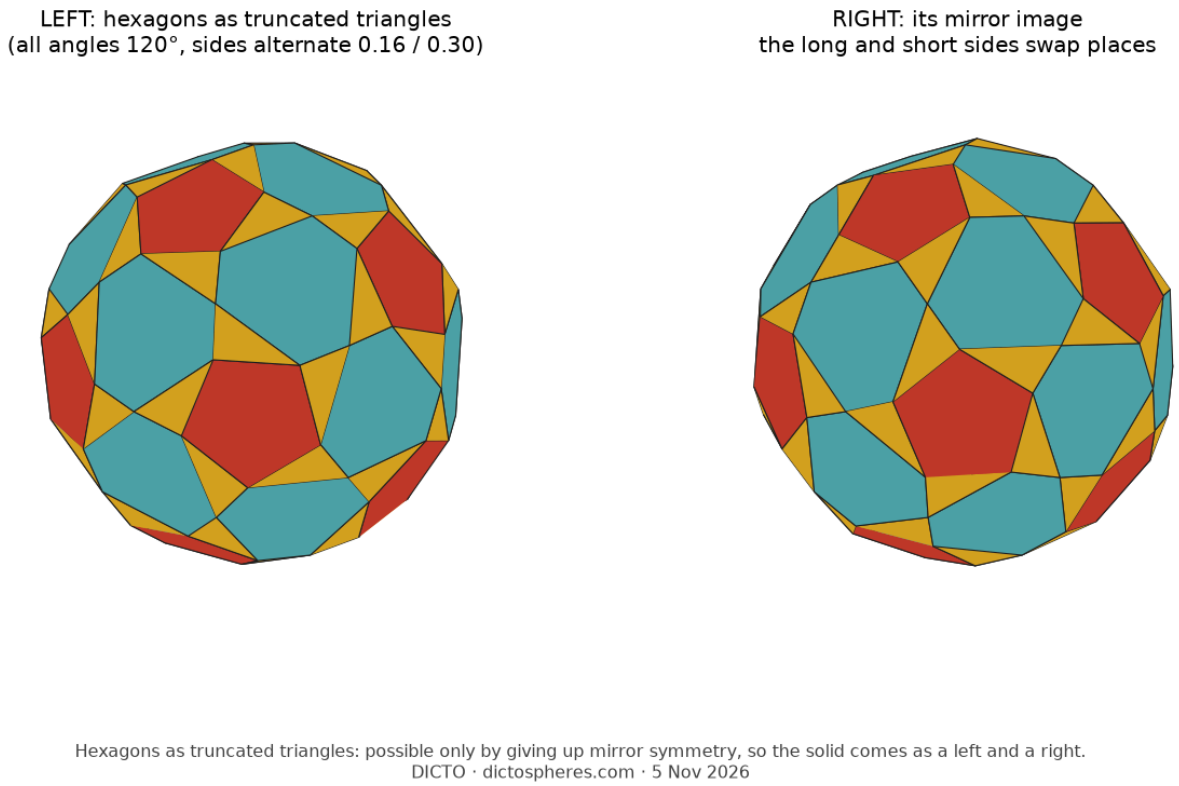


Figure 3. The hexagons as truncated triangles (all angles 120°, sides 0.16 and 0.30): the left and right forms.

4. Where the OG lives: the octet truss and the Bull-horns network

The octet truss. The OG sits in every octahedron of the octet truss with its eight joint hexagons on the faces (DICTO DOI: 10.5281/zenodo.23457016); the cut solid does too, since the hexagons are unchanged. Across every octahedron edge two copies now face each other with parallel pentagons, turned 36° relative to each other, 0.206 apart and offset sideways by 0.350; their convex hull, a slanted pentagonal antiprism, is the connector. DICTO proposed adapting the octet kit; with the connector it closes with four shapes and one mirror pair (float, manifold booleans; Figures 4 and 5): the cut solid (volume 1.132872, one per octahedron); the connector (0.018887, six per octahedron, two thirds inside the octahedra); the corner piece (0.07826, six per octahedron, all one shape, no left and right); and the dented tetrahedron (0.397254), left in one class of tetrahedra and right in the other. Their volumes add up to an octahedron and two tetrahedra (2.510264), and no two pieces overlap.

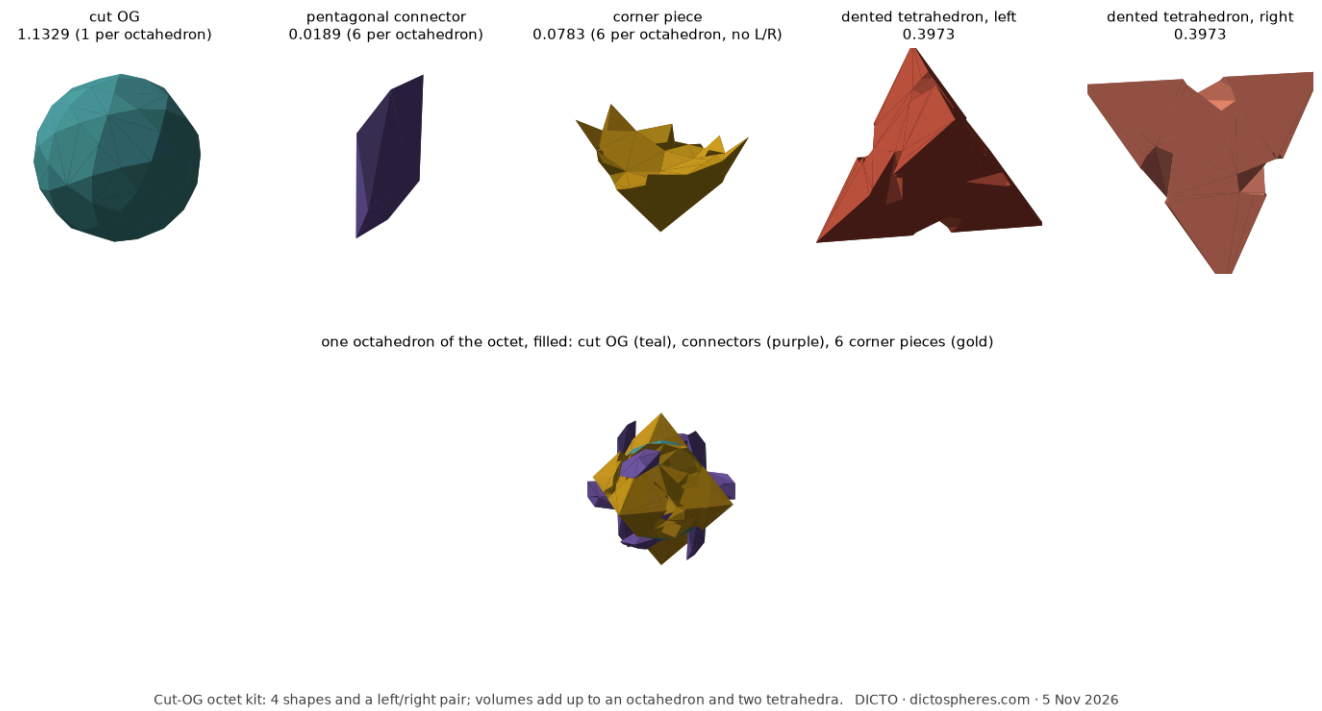


Figure 4. The cut-OG octet kit: the cut OG, the pentagonal connector, the corner piece, and the left and right dented tetrahedra.

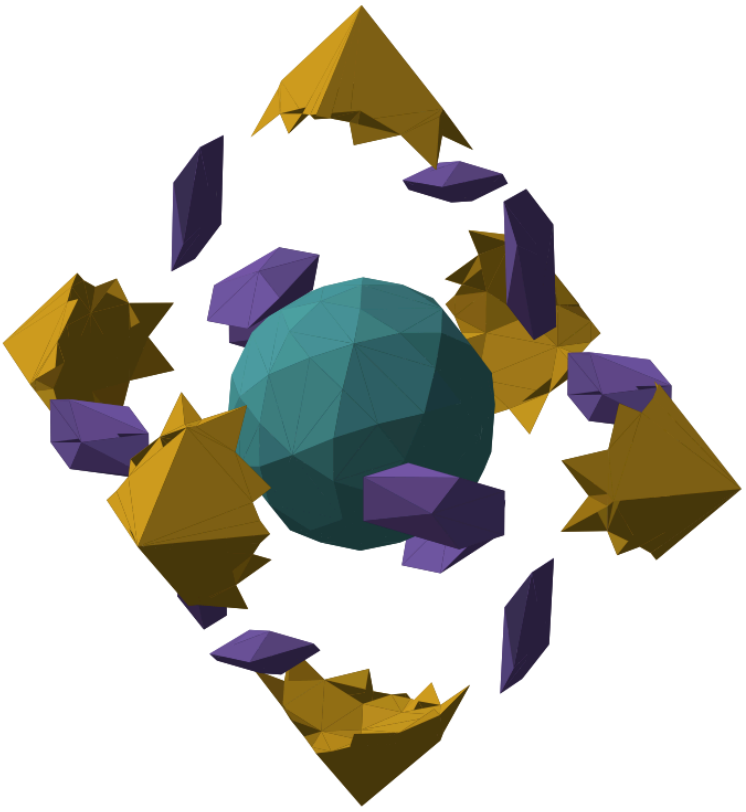


Figure 5. One octahedron, exploded: the cut OG, the twelve connectors on its edges (each shared with the neighbour across that edge) and the six corner pieces.

DICTO's Bull-horns adaptor. In the body-centred cubic double network, DICTO's Bull-horns OG adaptor fills the gap with one piece per hole between six copies (DICTO DOI: 10.5281/zenodo.23451210). DICTO proposed the same for the cut solid. The same construction (the Voronoi cell of each hole among all holes, minus the six copies that bite it) gives a new adaptor of volume 0.118230, exactly one third of a copy's gap: one solid without holes, centrally symmetric with a mirror plane, sphericity 0.514 (0.502 before). The cut solid and its adaptor fill space with two shapes, 76.16% and 23.84%. Where the twelve tips bit

small notches into the old adaptor, the new one meets the pentagons with flat faces (Figure 6). Run on the OG itself, the same program returns the published adaptor volume 0.111898.

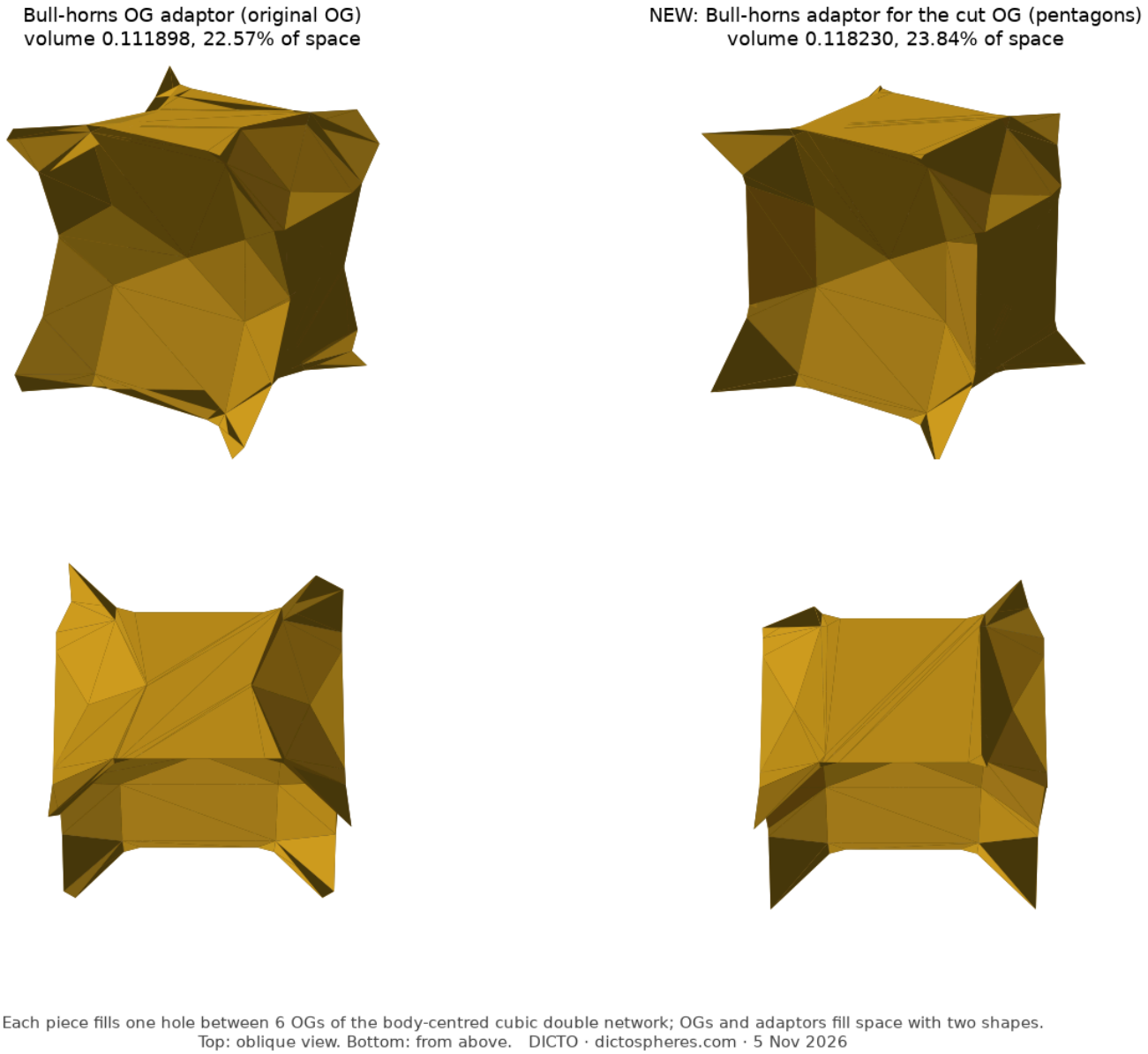


Figure 6. DICTO's Bull-horns OG adaptor (left) and the adaptor for the cut solid (right); top, oblique; bottom, from above.

5. What it did not settle, and what it opened

Not settled. The Dehn invariant exactly (only the float screen); a closed form for the all-equal-edge solid; whether that solid appears in the near-miss catalogues as it is; the twin layer of the gyrated octet: connectors fit on all 230 shared edges of a test patch with no overlap, but the pieces left at the twin layer are not yet identified.

Opened. (1) The corners of the cut solid sit at the bond centres of C_{60}, and solid C_{60} stacks its molecules face-centred cubic (Heiney et al. 1991), the close packing of DICTO's other work; we note the link and do not pursue it. (2) The truncated-triangle family is chiral, and in the gyrated octet the layers meet as mirror twins: a left and a right meeting at a twin layer is the same pattern as DICTO's self-mirroring Narcissus, and both bear on whether one piece can fill every stacking. (3) The pentagonal connector is a new kind of joint for OG networks.

6. Shapes on the way

Shape	Status
Cut OG on one sphere (20 hexagons, 12 pentagons, 60 triangles)	Rediscovered: the rectified truncated icosahedron; its relation to the OG is new in the sources checked
All-equal-edge version	Arrangement known among near-misses; this solid to be checked
Truncated-triangle family, left and right	Not found in the sources checked
Pentagonal connector (slanted pentagonal antiprism)	New as a piece; unnamed
Corner piece of the octet kit; dented tetrahedra, left and right	New pieces; unnamed
Bull-horns adaptor for the cut OG	New; derived from DICTO's Bull-horns OG adaptor

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